

## Basic Functional Test Cases

| Test Case                         | Input                   | Expected Output                                           |
|-----------------------------------|-------------------------|-----------------------------------------------------------|
| 1. Search for existing ZIP code   | -z10001                 | Full record printed for 10001                             |
| 2. Search for multiple valid ZIPs | -z10001 -z90210 -z33101 | 3 valid records displayed                                 |
| 3. Search for non-existent ZIP    | -z99999                 | "Zip Code 99999 not found in the file."                   |
| 4. Mix of valid and invalid       | -z10001 -z99999         | 1 record printed, 1 not found                             |
| 5. No ZIP codes provided          | none, just filename     | Program enters interactive mode                           |
| 6. ZIP flag but no file           | -z10001                 | Error: Please specify the filename before zip code flags. |

## Edge Case Test Cases

| Test Case                 | Input           | Expected Output                                                                  |
|---------------------------|-----------------|----------------------------------------------------------------------------------|
| 7. ZIP code with letters  | -zABCDE         | Error: invalid zip code argument                                                 |
| 8. Negative ZIP code      | -z-12345        | Error: invalid zip code argument                                                 |
| 9. Very large number      | -z999999999     | Likely not found or overflow behavior                                            |
| 10. Duplicate ZIP codes   | -z10001 -z10001 | Same record printed twice                                                        |
| 11. ZIP with leading zero | -z00501         | Correct record shown if present (ZIP codes are parsed as int, leading 0 dropped) |